

Hippocampal-Neocortical Interaction: A Hierarchy of Associativity

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ABSTRACT: The structures forming the medial temporal lobe appear to be necessary for the establishment of long-term declarative memory. In particular, they may be involved in the “consolidation” of information in higher-order associational cortices, perhaps through feedback projections. This review highlights the fact that the medial temporal lobe is organized as a hierarchy of associational networks. Indeed, associational connections within the perirhinal, parahippocampal, and entorhinal cortices enables a significant amount of integration of unimodal and polymodal inputs, so that only highly integrated information reaches the remainder of the hippocampal formation. The feedback efferent projections from the perirhinal and parahippocampal cortices to the neocortex largely reciprocate the afferent projections from the neocortex to these areas. There are, however, noticeable differences in the degree of reciprocity of connections between the perirhinal and parahippocampal cortices and certain areas of the neocortex, in particular in the frontal and temporal lobes. These observations are particularly important for models of hippocampal-neocortical interaction and long-term storage of information in the neocortex. Furthermore, recent functional studies suggest that the perirhinal and parahippocampal cortices are more than interfaces for communication between the neocortex and the hippocampal formation. These structures participate actively in memory processes, but the precise role they play in the service of memory or other cognitive functions is currently unclear. *Hippocampus* 2000;10:420–430.

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INTRODUCTION: MEDIAL TEMPORAL LOBE AND MEMORY

Beginning with the description of the amnesic patient H.M. by Scoville and Milner (1957), a variety of lines of evidence have clearly established the importance of the medial temporal lobe to memory function. Recent advances in characterizing human memory processes indicate that the brain supports multiple memory systems, and that these distinct memory systems are subserved by different neural systems (Squire, 1992). In humans, damage to the medial temporal lobe results in a severe memory impairment that is

characterized by rapid forgetting of declarative information but leaves intact a rather disparate set of memories grouped under the term “nondeclarative memory,” including procedural learning, priming, nonassociative learning, and simple classical conditioning (Squire and Zola-Morgan, 1996).

The structures forming the medial temporal lobe, including the dentate gyrus, hippocampus, subiculum, pre-subiculum, parasubiculum, and entorhinal, perirhinal, and parahippocampal cortices (Fig. 1), appear to be necessary for the establishment of long-term declarative memory in human and nonhuman mammals (Squire and Zola-Morgan, 1996). The structures of the medial temporal lobe are thought to be involved in the “consolidation” of information in higher-order associational cortices via feedback projections to these cortical areas, thus enabling them to serve as the final repositories for long-term memories (Squire and Alvarez, 1995). Consistent with this hypothesis, temporally graded retrograde amnesia has been demonstrated both in humans and animals after selective lesions of the medial temporal lobe (Scoville and Milner, 1957; Cho et al., 1995; Myhrer and Wangen, 1996). This model suggests a temporary role for the medial temporal lobe circuitry in the establishment of long-term memory in the neocortex (Squire, 1992). In this model, information initially processed in the neocortex is transmitted to the hippocampal complex through the different structures of the medial temporal lobe, i.e., the entorhinal, perirhinal, and parahippocampal cortices. This information is further processed within the remainder of the hippocampal formation, and the resultant “representation” is returned to the regions of origin, enabling the consolidation of long-term memory in the neocortex. The hippocampal circuitry may contribute to the formation of new relationships in long-term memory, such as those formed when associating a stimulus and its spatial/temporal context (thus representing a new event or episode), or when associating facts and the semantic context to which they belong (thus representing a new concept). This model, however, does not delineate more specifically the possible roles of the different structures of the medial temporal lobe in information processing.

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The development of animal models has been invaluable to our understanding of the function of the medial temporal lobe in memory processes. Animal studies have begun to contribute important information on the precise roles of each of the medial temporal lobe regions in the process of normal memory function. Studies carried out mainly in the monkey and the rat show that lesions restricted to the perirhinal or the parahippocampal cortices, for instance, produce a memory deficit as severe as lesions restricted to the hippocampus (Zola-Morgan et al., 1989; Otto and Eichenbaum, 1992; Meunier et al., 1993; Mumby and Pinel, 1994) and can also exacerbate the deficit observed after lesion of the hippocampus (Zola-Morgan et al., 1993; Wiig and Bilkey, 1995). Similarly, animal models have shown that the role of the entorhinal cortex is distinct from that of the hippocampus proper (Schenk and Morris, 1985; Meunier et al., 1993; Leonard et al., 1995). Recently, electrophysiological studies have also demonstrated the importance of the perirhinal and parahippocampal cortices in memory consolidation processes in animal models. In rats, perirhinal cortex activation appears to promote enhancement in cortico-cortical pathways secondarily activated by perirhinal inputs (Ivanco et al., 1996). In monkeys, Higuchi and Miyashita (1996) demonstrated that lesions of the entorhinal and perirhinal cortices prevented the formation of mnemonic neuronal responses to visual paired associates in the inferotemporal cortex.

It is now clear that these extrahippocampal structures do not simply funnel information from neocortical or subcortical areas to the hippocampus, but rather participate actively in memory processing. Unfortunately, however, little is known about the specific processing that takes place at each stage along the neocortical-hippocampal loop. We briefly describe the connectivity of these structures, in particular the connections between the neocortex, perirhinal, parahippocampal, and entorhinal cortices, as well as the important networks of intrinsic, associational connections within these structures.

Our presentation will be organized around several critical questions regarding the neuroanatomy of the medial temporal memory system:

1. What cortical regions send information to the hippocampal formation?
2. What are the neuroanatomical features that characterize the structures forming the neocortical-hippocampal loop?
3. How are the local connections organized in each of these regions and what are the implications for the types of information that ultimately reaches the hippocampal formation?
4. To what extent are the projections to the medial temporal lobe reciprocated?

NEUROANATOMY OF THE ENTORHINAL, PERIRHINAL, AND PARAHIPPOCAMPAL CORTICES

The entorhinal, perirhinal, and parahippocampal cortices surround the rhinal sulcus on the ventromedial surface of the monkey

brain (Figs. 1, 2). The terminology for the various subdivisions of these regions of cortex has been described previously (Amaral et al., 1987; Insausti et al., 1987). Briefly, the entorhinal cortex is subdivided into six parts that are arranged from rostral to caudal. Rostrally and medially is the olfactory division (EO), so named because it is the only entorhinal field that receives a direct projection from the olfactory bulb. EO is bordered laterally by the rostral subdivision (ER) of the entorhinal cortex. Area ER, in turn, is bordered laterally by the lateral subdivision of the entorhinal cortex (EL). Both EO and ER are bordered caudally by the intermediate subdivision of the entorhinal cortex (EI). Finally the caudal portion of the entorhinal cortex is made up of the larger caudal division (EC), and a smaller caudal limiting division (ECL).

The perirhinal cortex consists of a smaller medially situated area 35 and a larger laterally situated area 36. For most of its rostrocaudal extent, area 35 is confined to the fundus and lateral bank of the rhinal sulcus. Area 36 is located just lateral to area 35. At the most rostral and dorsal extent of the perirhinal cortex is area 36d, which comprises approximately the dorsal one fourth of what is typically referred to as temporal pole. Caudally adjacent to area 36d is area 36r (further subdivided into 36rm and 36rl). It is bounded laterally by the unimodal visual area TE. The caudal extreme of the perirhinal cortex has been called area 36c (further subdivided into 36cm and 36cl). Areas 36cm and 36cl are medially adjacent to the intermediate and caudal divisions of the entorhinal cortex (areas EI and EC), and are typically bounded laterally by the most rostral portion of area TF of the parahippocampal cortex.

The parahippocampal cortex is caudally adjacent to the perirhinal cortex and consists of a smaller, medially situated area TH and a larger laterally situated area TF (Von Bonin and Bailey, 1947), which is comprised of two subdivisions, a medially located area TFm and a laterally adjacent area TFl (Suzuki and Amaral, 1994a; Kobayashi and Amaral, 1999).

Connectivity

Neuroanatomical studies have provided detailed information about the interconnections between the structures of the medial temporal lobe and other cortical areas (Figs. 2, 3). A prominent feature of the perirhinal and parahippocampal cortices is their connectivity with the hippocampal formation, particularly the entorhinal cortex. In the monkey, these cortical regions provide almost two thirds of the neocortical input to the entorhinal cortex (Insausti et al., 1987), which in turn provides the predominant cortical input to the dentate gyrus and hippocampus via the perforant path (Amaral et al., 1987, Van Hoesen and Pandya, 1975; Witter et al., 1989; Suzuki and Amaral, 1990; Witter and Amaral, 1991). The functional significance of the perirhinal and parahippocampal projections to the entorhinal cortex is tied to the fact that the perirhinal and parahippocampal cortices are themselves the focus of converging inputs from unimodal and polymodal associational cortices (Jones and Powell, 1970; Van Hoesen and Pandya, 1975, Van Hoesen et al., 1975, Suzuki and Amaral, 1994a). The perirhinal and parahippocampal cortices are also in receipt of efferent projections from the entorhinal cortex that generally reciprocate the input pathways (Suzuki and Amaral, 1994b), and in turn

project back to the neocortex (Van Hoesen, 1982; Lavenex et al., 1998).

Another important and hitherto underappreciated feature of these cortices is the important network of intrinsic associational connections at every level of the neocortical-hippocampal loop. The intrinsic connections within the perirhinal and parahippocampal cortices potentially lead to substantial integration of information that reaches different rostrocaudal and mediolateral regions of these cortices. Moreover, the connections between the perirhinal and parahippocampal cortices provide another level of associativity and integration of information. Information processed within these cortices is then transmitted to the entorhinal cortex, the second stage of the neocortical-hippocampal loop that represents a hierarchically higher level of associativity, where the important network of intrinsic connections further increases the level of integration of information before it is transmitted to the hippocampus. The remaining part of the hippocampal formation, with its several associational networks within the dentate gyrus, CA3, and subiculum, constitutes the ultimate level of integration of information.

It is this hierarchy of associativity that most uniquely characterizes the neuroanatomy of the medial temporal lobe. We will provide a brief overview of these connections and then raise some possible implications of this type of convergent/divergent pattern of connections.

Cortical Afferent Projections to the Perirhinal and Parahippocampal Cortices

The perirhinal and parahippocampal cortices receive convergent inputs from a variety of unimodal and polymodal association cortices in the temporal, frontal, and parietal lobes (Fig. 2). Unimodal associational inputs arise from higher-order somatosensory, auditory, and visual association cortices (Van Hoesen and Pandya, 1975; Van Hoesen et al., 1975; Tranel et al., 1988; Suzuki and Amaral, 1994a). While the majority of the cortical inputs to the perirhinal and parahippocampal cortices are from visual areas, different types of visual information reach each of these cortices. Visual object information from area TE reaches predominantly the perirhinal cortex, whereas visuospatial information from area V4 and the posterior parietal cortex is directed more specifically to the parahippocampal cortex. Cortices associated with somatosensory processing project to both the perirhinal and parahippocampal cortices, and the auditory association cortex projects mainly to the parahippocampal cortex (Tranel et al., 1988; Suzuki and Amaral, 1994a). The perirhinal and parahippocampal cortices receive polysensory information from the ventrolateral and orbitofrontal cortices, cingulate and retrosplenial cortices, posterior parietal cortex, and polymodal region of the dorsal bank of the superior temporal sulcus (Suzuki and Amaral, 1994a).

Intrinsic Connections Within the Perirhinal and Parahippocampal Cortices

Neuroanatomical tracer studies have revealed a high level of associational connections within the perirhinal and parahip-

pocampal cortices (Suzuki and Amaral, 1994b; Lavenex and Amaral, unpublished observations).

Areas 36r and 36c of the perirhinal cortex are highly and reciprocally interconnected, whereas the most dorsal part of the perirhinal cortex, area 36d, seems to have fewer associational connections with the rest of the perirhinal cortex. The most rostral and medial part of area 36r of the perirhinal cortex projects heavily to itself, areas 35 and 36c, but has practically no projections directed to area 36d. Interestingly, the more lateral aspects of area 36r send heavy projections to areas 35, 36r, and 36c, as well as light projections to area 36d. The caudal area of the perirhinal cortex, area 36c, projects heavily to itself and moderately to heavily to area 36r; there are practically no projections from area 36c to area 36d.

Areas TF and TH are strongly and reciprocally interconnected. There does not seem to be any rostrocaudal organization of the projections, as the most rostral and caudal parts of area TF project heavily to the entire rostrocaudal axis of the parahippocampal cortex (area TF and area TH). In contrast, there seems to be some segregation of the projections along the mediolateral axis. Indeed, area TH projects heavily to itself and the medial portion of area TF, whereas the projections to the lateral portion of area TF are only moderate. Similarly, the lateral area seems to project heavily to itself, moderately to medial area TF, and only lightly to area TH. Projections originating at the border between the medial and lateral TF project strongly to all mediolateral portions of TF and only moderately to area TH.

These networks of intrinsic associational connections within the perirhinal and parahippocampal cortices potentially lead to the substantial integration of information that reaches different rostrocaudal and mediolateral regions of these cortices.

Reciprocal Connections Between the Perirhinal and Parahippocampal Cortices

As for the intrinsic connections within the perirhinal and parahippocampal cortices, neuroanatomical tracer studies have revealed an important network of reciprocal, associational connections between the perirhinal and parahippocampal cortices (Suzuki and Amaral, 1994b; Lavenex and Amaral, unpublished observations).

As stated previously, a prominent input to the perirhinal cortex arises from the parahippocampal cortex. Retrograde tracer injection studies show that areas 35, 36r, and 36c of the perirhinal cortex receive moderate to heavy projections from the rostral part of areas TF and TH, whereas more caudal portions of areas TF and TH project only lightly to areas 35, 36r, and 36c of the perirhinal cortex. In contrast, the most dorsal part of the perirhinal cortex, area 36d, receives a prominent projection from the entire rostrocaudal extent of the parahippocampal cortex. Anterograde tracer injection studies confirm that the most rostral part of area TF projects heavily to the most dorsal and the more caudal regions of the perirhinal cortex, areas 36d and 36c, whereas the projections to areas 35 and 36r are rather sparse. The mid-rostrocaudal region of area TF projects moderately to heavily to area 36d, lightly to moderately to area 36c, and sparsely to areas 35 and 36r. The caudal part of area TF sends moderate to heavy projections to areas 36d,

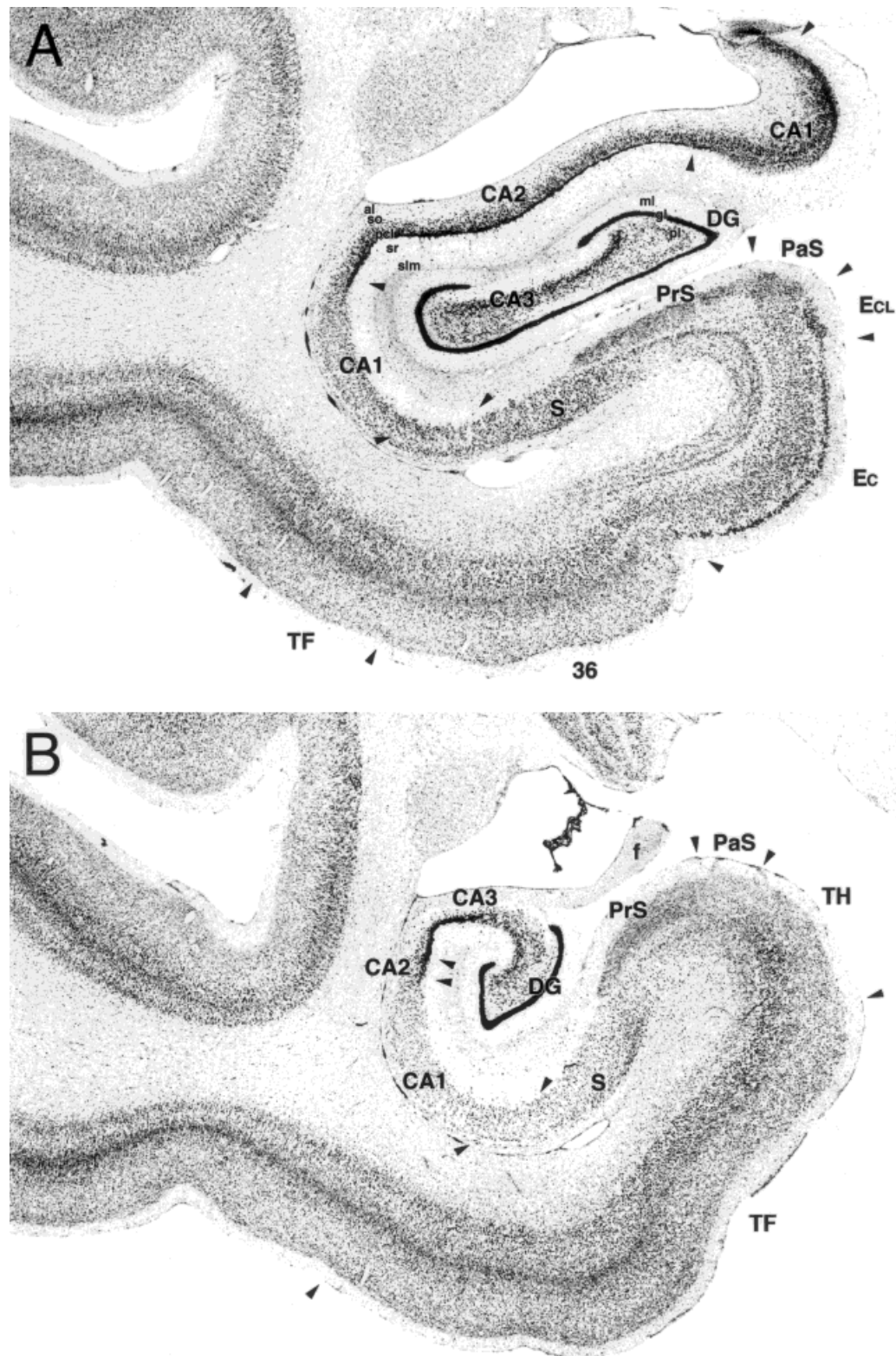


FIGURE 1. Coronal Nissl-stained sections through the medial temporal lobe of the macaque monkey. 36, area 36 of the perirhinal cortex (area 35 not shown); CA1, CA2, CA3, fields of the hippocampus; al, alveus; pcl, pyramidal cell layer; so, stratum oriens; slm, stratum lacunosum moleculare; DG, dentate gyrus; gl, granule-cell layer;

ml, molecular layer; pl, polymorphic cell layer; EC, entorhinal cortex, caudal division; ECL, entorhinal cortex, caudal limiting division; F, fimbria; PaS, parasubiculum; PrS, presubiculum; S, subiculum; TF, area TF of the parahippocampal cortex; TH, area TH of the parahippocampal cortex.

whereas there are practically no projections to areas 35, 36r, and 36c.

Retrograde tracer injection studies indicate that areas TF and TH receive light to moderate projections from areas 36d and 36c of the perirhinal cortex, whereas projections from areas 35 and 36r to the parahippocampal cortex are very sparse. Anterograde tracer injection studies confirm that there are practically no projections from the rostral and medial part of area 36r of the perirhinal cortex to areas TF and TH of the parahippocampal cortex, whereas a more lateral aspect of area 36r sends light to medium projections to the more rostral part of area TF. The caudal area of the perirhinal cortex, area 36c, projects moderately to heavily to the rostral portion of area TF. There are practically no projections from area 36c to the caudal portion of area TF or area TH.

These patterns of connectivity both within and between the perirhinal and parahippocampal cortices are particularly important when considering the topography of sensory input to this region and the level of integration of information at this stage of the neocortical-hippocampal loop. Indeed, information carried along some cortical inputs that do not project directly to the perirhinal cortex (e.g., area V4 and posterior parietal cortex) may eventually reach the perirhinal cortex through a disynaptic connection via the parahippocampal cortex.

Cortical Afferent Projections to the Entorhinal Cortex

The perirhinal and parahippocampal cortices, taken together, constitute the major sources of cortical input to the monkey entorhinal cortex, the second stage of the neocortical-hippocampal loop. Their patterns of connection differ significantly, however, with the perirhinal cortex projections terminating in the rostral two thirds of the entorhinal cortex, and the parahippocampal cortex projections terminating in the caudal two thirds (Suzuki and Amaral, 1994b). It is important to note, however, that this segregation of projections does not necessarily reflect a segregation of sensory inputs within the entorhinal cortex. Visuospatial information from area V4 and the posterior parietal cortex, for instance, can potentially reach every level of the entorhinal cortex, either through a disynaptic connection via the parahippocampal cortex or a trisynaptic connection via projections to the parahippocampal cortex relayed through the perirhinal cortex. The perirhinal and parahippocampal cortices are not the only cortical regions that project to the entorhinal cortex (Fig. 2). It also receives prominent projections from the superior temporal gyrus, the insular cortex, the orbitofrontal cortex, the cingulate cortex, and the retrosplenial cortex (Insausti et al., 1987). As with the perirhinal and parahippocampal cortices, each of these cortical regions receives rather diverse neocortical inputs, and each of these regions can be classified as "polysensory associational cortices" (Pandya and Yeterian, 1985). Thus, while the hippocampal formation appears to receive direct inputs from a relatively limited number of neocortical regions, each of these regions is a convergence zone that likely provides information from a much broader extent of cortex. In this way, the hippocampal formation is ultimately linked to much of the processing that takes place within the neocortex.

Intrinsic Connections Within the Entorhinal Cortex

The entorhinal cortex constitutes the second stage of the neocortical-hippocampal loop that represents a hierarchically higher level of associativity than the polysensory cortices that provide its input. Within the entorhinal cortex a major network of intrinsic, associational connections further increases the level of integration of information before it is transmitted to the hippocampus. The dominant organizational feature of intrinsic connections within the entorhinal cortex is the relative segregation of projections into three rostrocaudally oriented bands (Chrobak and Amaral, 1997): a lateral band that encompasses field EL and the most lateral aspect of fields EC and ECL; a wider, intermediate band that subsumes much of fields ER, EI, EC, and ECL; and a rostromedial band that includes EO, and the most medial aspect of EI. These data confirm earlier work carried out in the rat that entorhinal neurons are organized into three separate associational networks that may serve as a substrate for parallel processing within the hippocampal formation (Dolorfo and Amaral, 1998). A comparison with previously described connectivity of the rat entorhinal cortex, however, indicates clear differences between species. In the rodent entorhinal cortex, the distribution of projections from any location extends the entire rostrocaudal extent of the entorhinal cortex. This is not the case in the monkey, where the distribution of projections from any location extends about half the rostrocaudal extent of the entorhinal cortex. Thus, there appears to be greater segregation of connectivity domains within the entorhinal cortex and with respect to its interconnections with the hippocampus in the monkey as compared to the rat. This higher degree of segregation might indicate a higher level of parallel processing in the entorhinal cortex of the monkey than in the rat.

The associational connections in the entorhinal cortex originate both in the deep layers (V and VI) and in layer II. The rostrocaudal organization and the total extent of projections within both the deep and superficial layers were unrelated to whether cells originating these projections were located in the deep or superficial layers (Chrobak and Amaral, unpublished observations). However, whereas the deep cells projected both to other deep cells as well as to superficial layers, the layer II associational connection was maintained primarily within layer II. Thus, the layer II cells which generate the perforant path projection to the dentate gyrus and hippocampus are themselves interconnected by a dense network of associational connections. Moreover, the associational connections from deep to superficial layers have the potential of closing the hippocampal loop, since the output of the hippocampus and subiculum to the entorhinal cortex terminates mainly in the deep layers.

Hippocampal Circuitry

The connectional organization of the hippocampus has been extensively reviewed (e.g., Swanson et al., 1987; Amaral and Insausti, 1990; Kobayashi and Amaral, 1999), so we will only emphasize the critical features pertaining to the organization of the hierarchy of associativity along the neocortical-hippocampal loop.

The entorhinal cortex projects via the perforant path to the dentate gyrus, hippocampus, and subiculum. The projections to the dentate gyrus and CA3 field of the hippocampus originate mainly from cells in layer II. The projections to CA1 and subiculum, in contrast, originate mainly from cells in layer III. The dentate gyrus gives rise to only one major connection within the hippocampal formation. The axons of the granule cells, the mossy fibers, project to the CA3 region and also send collaterals which terminate on cells within the polymorphic layer of the dentate gyrus. The cells of the polymorphic layer, and particularly one cell type called the mossy cell, give rise to extensive associational connections that terminate on the proximal dendrites of granule cells throughout much of the length of the dentate gyrus. Projections arising from the CA3 pyramidal cells include collaterals to other CA3 pyramidal cells that constitute an extensive system of associational connections, as well as the Schaffer collaterals that constitute the major projection to the CA1 pyramidal cells. Both the CA3-CA3 associational connections and the Schaffer collaterals diverge widely from their point of origin. A single CA3 pyramidal cell may contact CA1 cells throughout 75% of the length of the hippocampus. The CA1 field of the hippocampus projects both to the subiculum and to the deep layers of the entorhinal cortex. Unlike the CA3 field, however, there are very few associational connections within CA1. This fact distinguishes CA1 from most other regions of the medial temporal lobe.

Efferent Projections From the Entorhinal Cortex to Perirhinal and Parahippocampal Cortices

The perirhinal and parahippocampal cortices receive feedback efferent projections from the hippocampus and the entorhinal cortex that largely reciprocate the input pathways (Van Hoesen, 1982; Suzuki and Amaral, 1994b). Accordingly, the perirhinal and parahippocampal cortices play a unique role in the dissemination of hippocampal output. The degree of reciprocity of the interconnections of the entorhinal cortex with the perirhinal and parahippocampal cortices differs slightly. The parahippocampal/entorhinal connections have a high degree of reciprocity, so that the caudal entorhinal mainly projects to the parahippocampal cortex. In contrast, the degree of reciprocity of the perirhinal/entorhinal interconnections varies depending on the mediolateral position within the perirhinal cortex; medial portions of the perirhinal cortex exhibit a higher degree of reciprocity than the lateral portion. Overall, however, one can consider the efferent projections from the entorhinal cortex to the perirhinal and parahippocampal cortices as feedback projections that are largely reciprocal.

Efferent Projections From Perirhinal and Parahippocampal Cortices to the Neocortex

The perirhinal and parahippocampal cortices do not simply relay information from the neocortex to the hippocampus, but receive strong efferent projections from the hippocampal formation (see above), and in turn project back to the neocortex. The efferent projections from the perirhinal and parahippocampal cortices to the neocortex are indeed extremely widespread (Van Hoesen, 1982). Recent analyses (Lavenex et al., 1998; Lavenex and Amaral,

unpublished observations) have shown a relatively high degree of reciprocity of the connections between the perirhinal and parahippocampal cortices and the neocortex. The perirhinal and parahippocampal cortices receive convergent inputs from a variety of unimodal and polymodal association cortices in the temporal, frontal, and parietal lobes (see above), and generally appear to project back to the same unimodal and polymodal association cortices. This degree of reciprocity seems to correspond, at first, to the pattern of connectivity generally observed for other cortico-cortical projections (Felleman and Van Essen, 1991; Van Hoesen, 1982). The reciprocity of the connections, however, is not absolute. Indeed, the neocortical cells that originate projections to the perirhinal and parahippocampal cortices may not necessarily be the same cells in particular cortical regions that receive the feedback projections (Lavenex and Amaral, unpublished observations). Moreover, there are clear discrepancies in the reciprocity of the connections between the perirhinal and parahippocampal cortices and the neocortex. The efferent projections from the perirhinal cortex to the frontal cortex, for instance, are much more limited and reach fewer areas than those that send projections to the perirhinal cortex. Similarly, the afferent projections from the superior temporal sulcus arise from a much wider area and are much more important than the efferent projections from the perirhinal cortex to the superior temporal sulcus. In contrast, the perirhinal cortex receives relatively few projections from the caudal visual areas V4 and TEO, and appears to send more widespread projections to these areas. These findings have important implications for the models of interaction between the hippocampus and neocortex and long-term storage of information in the neocortex.

Laminar Organization

The laminar distribution of the projections from one cortical area to another has been previously used to define hierarchical relationships in the visual system (Felleman and Van Essen, 1991). Cortico-cortical connections can be divided into three different categories, based on layers of origin and termination: feedforward (or ascending) projections, lateral (or local) projections, and feedback (or descending) projections (Rockland and Pandya, 1979). Despite modifications of the definitions necessitated by the periallocortical cytoarchitecture of the entorhinal cortex, the projections from the perirhinal and parahippocampal cortices to the entorhinal cortex appear to exhibit a feedforward (or ascending) type of organization. The projections from the perirhinal and parahippocampal cortices originate primarily from the superficial layer III and from layer V. Since the entorhinal cortex does not have a typical layer IV (internal granular layer), these feedforward projections terminate primarily in the more superficial layers II and III, that in turn project forward to the dentate gyrus and hippocampus. The organization of these feedforward (ascending) projections is thus typical of a hierarchical organization of these different cortical areas, with the perirhinal and parahippocampal cortices located at a lower level than the entorhinal cortex.

The efferent connections from the hippocampal formation to the neocortex indicate that these projections are typical of feedback (or descending) projections. Indeed, projections from CA1 and the

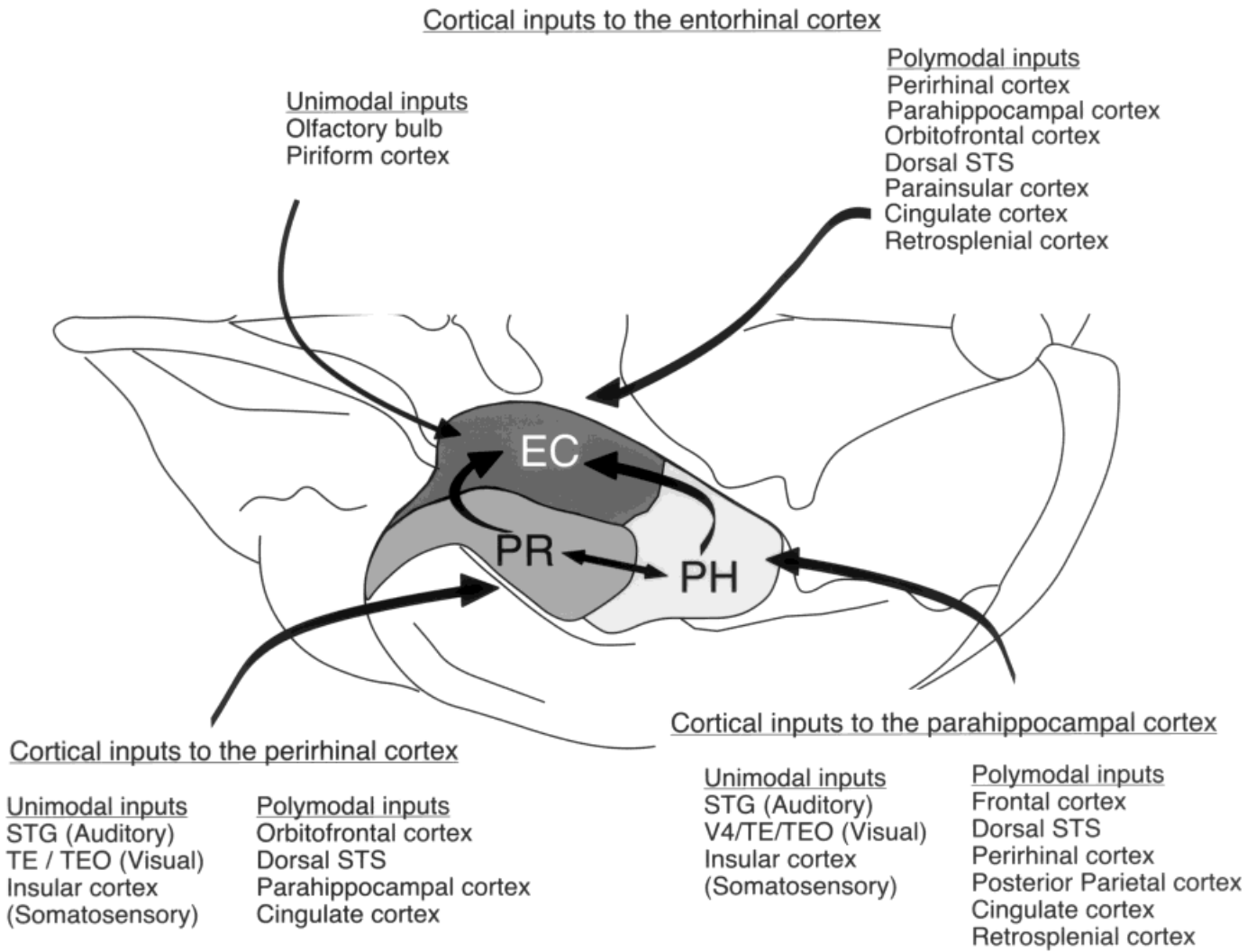


FIGURE 2. Summary diagram showing the ventral surface of the macaque monkey brain. The major connections of the entorhinal and perirhinal and parahippocampal cortices are indicated.

subiculum typically project to the deep layers of the entorhinal cortex. The projections from the entorhinal cortex to the perirhinal and parahippocampal cortices also follow a classic feedback pattern; the projections originate in layer V and terminate mainly in layer I and to a lesser extent in layers V/VI. The feedback projections from the perirhinal and parahippocampal cortices to the neocortex are also typical of feedback projections, originating mainly from the deep layers and terminating in the more superficial layers within the neocortex.

FLOW OF INFORMATION WITHIN THE NEOCORTICAL-HIPPOCAMPAL LOOP

The connectional anatomy described above tells us little about function itself. But it certainly helps to define the structural bound-

aries within which functional mechanisms must operate. Considering the neocortical-hippocampal loop, it is the hierarchical organization of associational networks that most uniquely characterizes the neuroanatomy of the structures forming the medial temporal lobe.

Previous models of the role of the hippocampus in memory consolidation generally considered the medial temporal lobe as a whole and neglected these “intermediate” cortices (i.e., the entorhinal, perirhinal, and parahippocampal cortices) that were used as mere relays to transfer information from the neocortex to the hippocampus and vice versa. Thus, schemes describing the flow of information through the medial temporal memory system usually represented information transmitted from the neocortex to the hippocampal complex and back out from the hippocampal complex to the neocortex. Functional studies, however, support the interpretation derived from the neuroanatomical data briefly described above, that these “intermediate” cortices do not act as mere

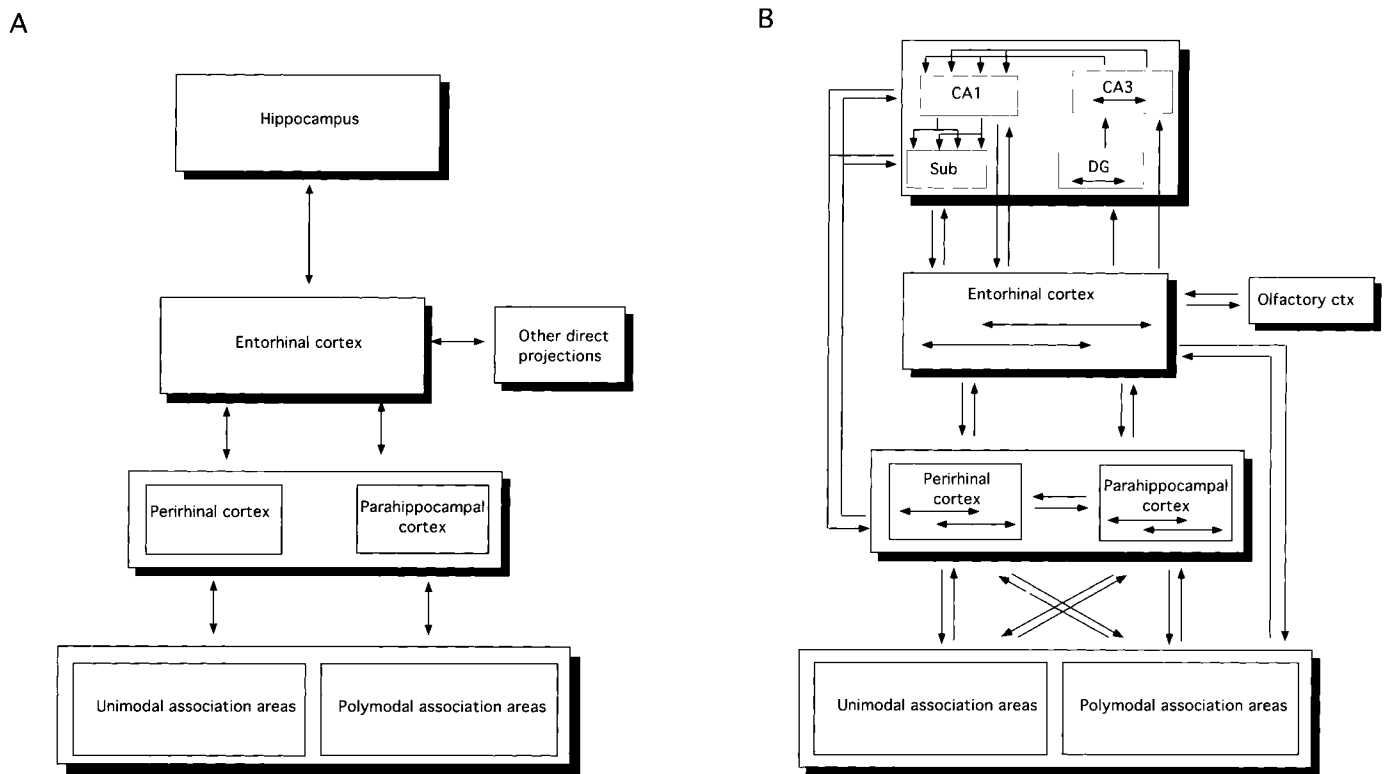


FIGURE 3. A: Schematic diagram, illustrating the standard view of projections within the medial temporal lobe. B: Updated version of the schematic diagram which emphasizes the associational networks

relays for the flow of information from the neocortex to the hippocampal complex, but participate actively in the memory processes.

These cortices form a hierarchy of associativity that increases in complexity when going from the neocortex to the hippocampal complex. Indeed, the associational connections within the perirhinal, parahippocampal, and entorhinal cortices and the convergence/divergence of inputs at different levels of the loop enable a significant amount of integration of unimodal and multimodal inputs well before information reaches the hippocampus. In terms of information processing, the level of integration/complexity of the information undoubtedly increases when moving from the neocortex to the hippocampal complex; unimodal information becomes polymodal, supermodal, or amodal, and eventually reaches the highest level of abstraction within the hippocampal complex before being sent back to the neocortex. The hippocampal output further influences the processing of incoming information through the feedback projections from the hippocampal complex to the neocortex and the associational connections within each structure (e.g., deep to superficial projections described in the entorhinal cortex). We will review the main characteristics of the medial temporal lobe memory system and try to suggest ways in which these characteristics might affect the role of these different structures in the integration of multisensory information and the consolidation of memory in the neocortex. This discussion generates two conclusions:

at each stage of the cortical-hippocampal loop. These projections tend to integrate information further while passing it along to the next stage in the hierarchy.

1. The integration, complexity, or abstraction of information likely increases as one progresses from the neocortex, to the perirhinal and parahippocampal cortex, to the entorhinal cortex and through the hippocampal circuit.
2. The output of the medial temporal lobe system is ultimately distributed, via feedback projections, to much of the neocortex, including unimodal sensory cortices. Thus, the higher-order representation formed by the medial temporal lobe, which is broadcast widely, may interact with lower level information processed or stored in unimodal cortices.

Perirhinal and Parahippocampal Cortices, the First Level of Integration

The perirhinal and parahippocampal cortices represent the first stage in the integration of multimodal polysensory information within the medial temporal lobe memory system. Early electrophysiological studies demonstrated that single neurons in the perirhinal and parahippocampal cortices are responsive to multimodal stimulation (Desimone and Gross, 1979). More recent electrophysiological studies have demonstrated the importance of the perirhinal cortex in the establishment of memory traces in the neocortex. In rats, perirhinal cortex activation appears to promote enhancement in cortico-cortical pathways secondarily activated by perirhinal inputs (Ivanco et al., 1996). In monkeys, Higuchi and Miyashita (1996) demonstrated the

role of the perirhinal cortex in the formation of mnemonic neuronal responses to visual paired associates in the inferotemporal cortex (TE). In their study, efferent projections from the entorhinal and/or perirhinal cortices were found to be necessary for the maintenance or consolidation of pair-coding information in area TE. Indeed, the ability of inferotemporal neurons to represent associations between picture pairs was lost after neurotoxin lesions of the entorhinal and perirhinal cortices, most likely through disruption of feedback neural signals to the inferotemporal neurons. Neuronal response patterns of neurons in the perirhinal cortex may extend well beyond sensory activation. In the delayed-matching-to-sample task, the pattern of activity of perirhinal cells during the delay (when no stimulus is present) reflects the memory of the sample stimulus (Miller et al., 1993; Miyashita and Chang, 1988), although the presentation of another nonmatching stimulus abolishes this delay activity (Miller et al., 1993). This suggests that the delay activity in perirhinal cells represents the neuronal response corresponding to the maintenance of an explicit representation of the stimulus held in memory (Miyashita and Chang, 1988).

A study of visual recognition recently showed that lesions of the perirhinal cortex produce a deficit nearly as severe as that found after entorhinal plus perirhinal cortex lesions, whereas ablations of the entorhinal cortex alone produce only a mild deficit (Meunier et al., 1993). Similarly, Buckley and Gaffan (1998) and Buckley et al. (1997) showed in different tasks that the function of the perirhinal cortex can be dissociated from the function of the other structures of the medial temporal lobe, although the exact nature of the tasks and thus the function of the perirhinal cortex remain a matter of debate.

Taken together, these data indicate that perirhinal neurons are multimodal and may be coding more than sensory attributes. The precise characterization of the response properties of perirhinal neurons requires substantial additional research. It is not clear, for example, if there are topographic differences in response attributes. It will also be important, albeit difficult, to determine how dependent the perirhinal cortex is on the entorhinal cortex and hippocampus for the function(s) that it carries out.

Entorhinal Cortex, the Second Level of Integration

The entorhinal cortex represents the second stage in the integration of multimodal polysensory information within the medial temporal lobe system. Electrophysiological studies have shown that visually responsive cells and stimulus-selective cells are less common in the entorhinal cortex than in the perirhinal cortex (Suzuki et al., 1997). This suggests that cells in the entorhinal cortex are less involved in the analysis and coding of object/sensory features than the perirhinal cortex. Given the hierarchical organization of associational connectivity described above, entorhinal neurons might respond solely to the conjoint activation by neurons from several cortical areas, i.e., conjunction of information from separate multimodal polysensory areas, and not to the unique activation by neurons from a single cortical area, i.e., a simple stimulus. However, cells in the entorhinal cortex do respond differen-

tially to different objects, suggesting that some object information is retained in the entorhinal cortex and is presumably passed on to subsequent structures within the hippocampal formation.

The Hippocampal Formation, the Ultimate Level of Integration

As the ultimate structure along the neocortical-hippocampal loop, the hippocampal complex participates in the highest level of integration of information. It receives only highly integrated supermodal or amodal information from the entorhinal, perirhinal, and parahippocampal cortices and sends back even more highly processed, abstract, multidimensional information to the structures that originated these projections. Others have suggested that it is only after reaching the hippocampus that the memory representation attains the full richness that characterizes declarative memory (Squire, 1992; Cohen and Eichenbaum, 1993).

Timing of Information Processing

Given the neuroanatomical characteristics of the structures forming the medial temporal lobe (hierarchical organization, high level of associativity, and reciprocity of connections), it might seem difficult to distinguish between the different functional properties of these structures. Given the extreme interconnectivity of these structures, how can one try to determine the type of information processing that takes place at one particular level, independently from the other structures that interact with it? A solution to this problem might be to consider the timing of neuronal responses within the network. Transfer of information from one stage to the next along the neocortical-hippocampal loop is necessarily associated with delays. As one progresses from the neocortex to the hippocampus, neuronal response patterns should increase in informational complexity. At the same time, feedback projections send back a more integrated signal to the areas that preceded them. Consequently, one might be able to see neurons, located at the early stages of the loop, conveying different types of information in their firing patterns, starting at different latencies. Such a hypothesis is derived from an experiment by Sugase et al. (1999), who recorded single neurons in the superior temporal sulcus of the monkey. They demonstrated that single neurons conveyed two different scales of facial information in their firing patterns, starting at different latencies. At an early time point, the informational content of the neuronal responses could differentiate categories of faces (monkey vs. human) but not individuals of the category. At a later time point, neuronal responses had sufficient information to differentiate individuals within a category. This differential pattern of firing was thought to be due to the interaction with higher-order cortical areas through feedback projections. If each different stage of the neocortical-hippocampal loop processes different levels of integration of information, time-delay might also permit feedback contributions to multiple levels of coding by the same neurons in the early stages of the loop.

Another important aspect of timing in the modulation of information processing by feedback projections is the presence of theta oscillation in the perirhinal cortex (as demonstrated in the cat; Collins et al., 1999). During recording in different conditions, a

high degree of correlation was found between perirhinal and entorhinal EEG activities. In particular, during walking and paradoxical sleep, multiple simultaneously recorded entorhinal and perirhinal sites displayed an oscillation in the theta range that was highly correlated. Although these experiments did not examine the origin of the perirhinal theta, the available evidence suggests that it is synaptically driven by the entorhinal cortex and/or subiculum. Indeed, the perirhinal cortex receives no inputs from the pacemaker of hippocampal and entorhinal theta, the medial septum, but it is the target of strong projections from the deep entorhinal layers which contain neurons that display rhythmic theta-related firing. The presence of coherent theta oscillations in the perirhinal cortex could be very important, given the crucial role that coincident neuronal activity is believed to play in various forms of synaptic plasticity.

CONCLUSIONS

The structures forming the medial temporal lobe, including the dentate gyrus, hippocampus, subiculum, presubiculum, parasubiculum, and entorhinal, perirhinal, and parahippocampal cortices, appear to be necessary for the establishment of long-term declarative memory. These structures are thought to be involved in the "consolidation" of information in higher-order associational cortices via feedback projections. The neocortical-hippocampal loop is most uniquely characterized by the fact that it is a hierarchical organization of associational networks. These cortices form a hierarchy of associativity that increases in complexity when going from the neocortex to the hippocampal complex. Indeed, associational connections within the perirhinal, parahippocampal, and entorhinal cortices and the convergence/divergence of inputs at different levels of the loop enable a significant amount of integration of polysensory and multimodal inputs well before information reaches the hippocampus.

The perirhinal and parahippocampal cortices receive convergent inputs from a variety of unimodal and polymodal association cortices in the temporal, frontal, and parietal lobes. The perirhinal and parahippocampal cortices represent the first stage of this hierarchy of associational networks participating in the integration of multimodal polysensory information within the medial temporal lobe memory system. Together, the perirhinal and parahippocampal cortices constitute the major sources of cortical input to the monkey entorhinal cortex, the second stage of the neocortical-hippocampal loop. The entorhinal cortex represents a hierarchically higher level of associativity, where the important network of intrinsic, associational connections further increases the level of integration of information before it is transmitted to the hippocampus. Thus, while the hippocampal formation appears to receive direct inputs from a relatively limited number of neocortical regions, each of these regions is a convergence zone that likely provides information from a much broader extent of cortex. In this way, the hippocampal formation is ultimately linked to much of the processing that takes place within the neocortex.

The connections between the neocortex and the structures forming the medial temporal lobe are generally reciprocal. The afferent connections from the neocortex to the hippocampal complex indicate that these projections are typical of feedforward (or ascending) projections. The efferent connections from the hippocampal complex to the neocortex indicate that these projections are typical of feedback (or descending) projections. The connections between the perirhinal and parahippocampal cortices and the hippocampal formation are highly reciprocal. Accordingly, the perirhinal and parahippocampal cortices play a unique role in the dissemination of hippocampal output. The connections between the perirhinal and parahippocampal cortices and the neocortex also show a relatively high degree of reciprocity. However, there may not be a precise point-to-point relationship in the reciprocity of these connections. Moreover, there are clear discrepancies in the reciprocity of the connections, in particular between the perirhinal cortex and some areas in the frontal and temporal lobes. Together, these findings have important implications for the models of interaction between the hippocampus and neocortex and the long-term storage of information in the neocortex. One implication of this system is that the level of integration or abstraction of information reaches its highest state in the hippocampal formation, and this high-level abstraction is then returned widely to the neocortex. Whether each recipient cortical region uses this return information in a similar way or perhaps has a mechanism to extract some information while discarding the remainder will be of fundamental interest to models of the medial temporal lobe in memory function.

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